

3 Answers

3.1 Problem 1: Conceptual Understanding and Model Interpretation (40 points)

1. The ROS coefficient, $\beta_1 = 0.035$, means that for one unit increase in ROS, the expected gene expression increases by 0.035 unit while holding Cell age and Treatment constant.

The Cell age coefficient, $\beta_2 = -0.12$, means that for one unit increase in Cell age, the expected gene expression decreases by 0.12 unit while holding ROS and Treatment constant.

The Treatment coefficient, $\beta_3 = 0.85$, means that treated cells are predicted to have gene expression 0.85 units higher than control cells with the same ROS level and cell age.

2. $R^2 = 0.48$ means that approximately 48% of the variation in observed gene expression across the observed 120 cells is explained by the predictors included in the linear regression model. The remaining approximately 52% of the variation is not explained by the model. This is clear from how the variation is decomposed and how R^2 is calculated.

- Sum-of-squares decomposition:

$$SS_{tot} = SS_{reg} + SS_{err}$$

where

$$SS_{tot} = \sum_{i=1}^n (y_i - \bar{y})^2$$

$$SS_{reg} = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2$$

$$SS_{err} = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

This Sum-of-squares decomposition corresponds to partitioning the variation in Y .

- R^2 computation:

$$R^2 = \frac{SS_{reg}}{SS_{tot}}$$

Based on the equations above, it is clear that R^2 does not mean that individual predictions are 48% accurate. R^2 describes the proportion of variation explained by the model at the dataset level and does not directly indicate the magnitude of individual prediction errors.

In contrast, Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE) summarize the magnitudes of prediction errors across a dataset because they are calculated as follows:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$
$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

For prediction accuracy for individual samples, the following can be used:

Measure	Formula	Interpretation
Residual	$y_i - \hat{y}_i$	Direction and magnitude of error
Absolute error	$ y_i - \hat{y}_i $	Magnitude of error
Squared error	$(y_i - \hat{y}_i)^2$	Penalizes large errors strongly
Relative absolute error	$\frac{ y_i - \hat{y}_i }{ y_i }$	Error relative to observed value

3. Statistical significance does not establish causation. A significant ROS coefficient indicates an association between ROS and gene expression after adjustment for the other predictors in the model. Causal interpretation could be invalid because of:

- unmeasured confounding,
- reverse causation, a situation where you think variable X causes variable Y , but in reality, variable Y is causing variable X .
- measurement error,
- selection bias,
- omitted variables,
- or the study design itself.

A randomized experiment or a carefully justified causal-inference framework would be needed for a stronger causal conclusion.

4. Major Ordinary Least Squares assumptions include:

Assumption	Meaning	Possible diagnostic
Linearity	Expected outcome is a linear function of predictors	Residual-vs-fitted plot
Independence	Errors are independent across observations	Examine study design; residual autocorrelation if appropriate
Homoscedasticity	Error variance is approximately constant	Residual-vs-fitted plot
Normality of residuals	Residuals are approximately normally distributed	Q-Q plot or histogram

A fifth useful consideration is the absence of severe multicollinearity among predictors, which can be examined using predictor correlations or variance inflation factors.

A funnel-shaped residual plot suggests heteroscedasticity, meaning that error variance changes with the fitted outcome. Consequences may include unreliable standard errors, confidence intervals, and hypothesis tests, although OLS coefficient estimates can remain unbiased under other assumptions.

One possible remedy is to transform the outcome, such as a log transformation.

5. If treatment substantially increases ROS, ROS and treatment will be correlated. This can make it difficult to separate the independent effects of ROS and treatment.

The ROS coefficient represents the estimated ROS association conditional on treatment status, rather than the total relationship between ROS and gene expression.

Strong correlation between the predictors can also increase the variance of coefficient estimates and produce unstable coefficients. This is a form of multicollinearity.

6. Adding 50 meaningless predictors would generally have the following effects:

Quantity	Expected Behavior
Training R^2	Increases or stays the same
Adjusted R^2	May decrease if predictors provide insufficient improvement
Training error	Decrease or stays the same
Testing error	Often increases due to overfitting

3.2 Problem 2: Linear regression with a real biological dataset (60 points)

Check the python jupyter notebook for this problem.